

Module 2: Transport Layer and Network Layer



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Transport Layer

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8.5 Quality of Service (QoS)

- The Internet was originally designed for **best-effort service** with no guarantee of predictable performance.
- Best-effort service is sufficient for delay-insensitive traffic such as file transfer and e-mail.
- Such traffic is called **elastic** or **available bit rate** traffic.
- Multimedia applications can generate **real-time traffic**, which is delay-sensitive.
- **Quality of Service (QoS)** is a set of techniques and mechanisms that guarantees predictable network performance to an application.

8.5.1 Data-Flow Characteristics

Four Traditional Flow Characteristics

- 1 Reliability
- 2 Delay
- 3 Jitter
- 4 Bandwidth

- Different applications have different sensitivity to these characteristics.
- QoS mechanisms must match network treatment to application requirements.

Reliability

- Ability to deliver packets safely to the destination.
- Packet or acknowledgment loss may require retransmission.
- More important for e-mail, file transfer, and Internet access.
- Less important for telephony or audio conferencing.

Delay

- Source-to-destination delay.
- Telephony, audio/video conferencing, and remote login need minimum delay.
- Delay is less important for file transfer and e-mail.

Jitter

- Variation in delay among packets of the same flow.
- Equal delays are acceptable for audio/video.
- Variable delays can disrupt real-time media.

0, 1, 2, 3 $\rightarrow$ 20, 21, 22, 23

Same delay = 20

Bandwidth

- Applications require different data rates.
- Video conferencing may need millions of bits per second.
- An e-mail may contain far fewer bits.

Sensitivity of Applications

Application	Reliability	Delay	Jitter	Bandwidth
FTP	High	Low	Low	Medium
HTTP	High	Medium	Low	Medium
Audio-on-demand	Low	Low	High	Medium
Video-on-demand	Low	Low	High	High
Voice over IP	Low	High	High	Low
Video over IP	Low	High	High	High

- High reliability → error checking.
- High delay sensitivity → priority treatment.
- High jitter sensitivity → controlled packet delay.
- High bandwidth requirement → sufficient bandwidth allocation.

8.5.2 Flow Classes

- Internet flow classes have not been formally defined in the text.
- A flow can be described by the required level of each QoS characteristic.
- Example: FTP needs high reliability and probably medium bandwidth; delay and jitter are not important.

Example 8.11 – ATM

ATM formally defines five service classes: CBR, VBR-NRT, VBR-RT, ABR, and UBR.

ATM Flow Classes – Example 8.11

- ① **CBR – Constant Bit Rate:** emulates circuit switching; sensitive to cell-delay variation. Examples: telephone, video conferencing, television.
- ② **VBR-NRT:** variable rate, non-real-time; example: multimedia e-mail.
- ③ **VBR-RT:** variable rate, real-time; for interactive compressed video sensitive to cell-delay variation.
- ④ **ABR:** rate-based flow control for data traffic such as file transfer and e-mail.
- ⑤ **UBR:** unspecified bit rate; includes other traffic and is widely used for TCP/IP.

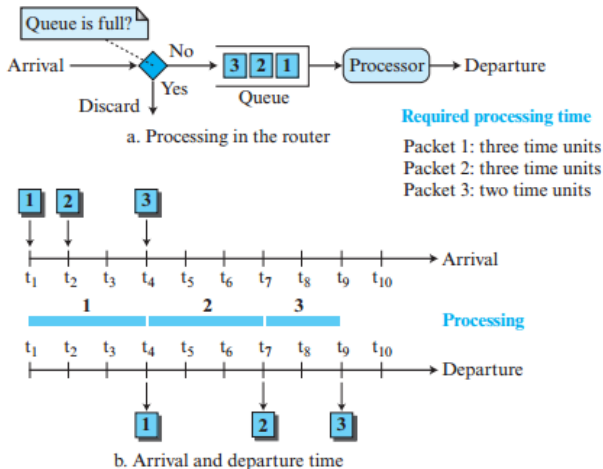
8.5.3 Flow Control to Improve QoS

- An IP datagram has a **ToS field** that can informally indicate required service.
- QoS problems mainly occur at routers, where packets can experience delay, jitter, loss, and bandwidth limitation.
- Scheduling techniques discussed:
 - ① FIFO queuing
 - ② Priority queuing
 - ③ Weighted fair queuing

- **FIFO = First-In, First-Out.**
- Packets wait in a buffer until the router is ready to process them.
- Packets depart in the order they arrive.
- If average arrival rate exceeds processing rate, the queue fills and new packets may be discarded.
- Different packet sizes can require different processing times and produce jitter.

Figure 8.59 – FIFO Queuing

Figure 8.59 *FIFO queue*



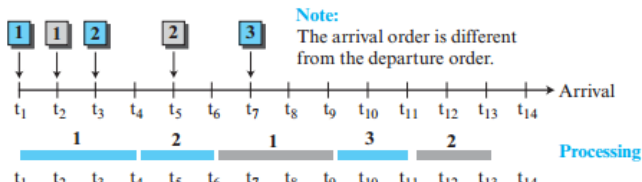
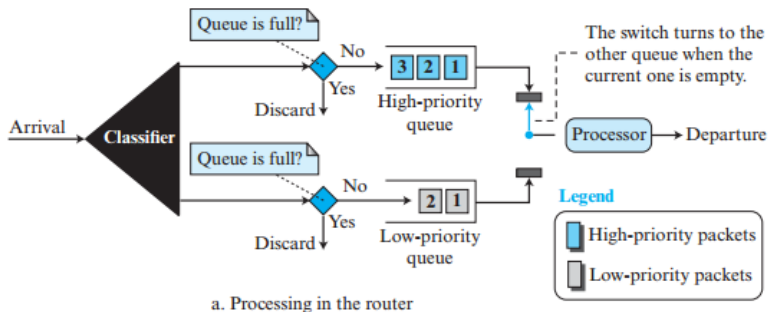
- FIFO is the default scheduling method described for the Internet.
- It provides no service differentiation between packet classes.
- FTP, Voice over IP, and e-mail packets are treated alike.
- Loss, delay, and jitter depend on traffic arriving at the router.
- Other scheduling mechanisms are needed for differentiated service.

Priority Queuing

- Packets are assigned to **priority classes**; each class has its own queue.
- Highest-priority queue is processed first.
- A queue is served until empty before moving to a lower-priority queue.
- Priority may be determined from IPv4 ToS, IPv6 priority, destination-address priority, or application/destination-port priority.

Figure 8.60 – Priority Queuing

Figure 8.60 Priority queuing



Priority Queuing – Benefit and Problem

Benefit

- Higher-priority traffic can reach the destination with less delay.
- Useful for multimedia and delay-sensitive traffic.

Problem: Starvation

- Continuous high-priority traffic may prevent lower-priority queues from being served.
- Severe starvation may result in packet loss.

Weighted Fair Queuing (WFQ)

- Packets are assigned to classes and placed in separate queues.
- Each queue receives a **weight** based on priority.
- Queues are served in round-robin fashion.
- Number of packets selected depends on the queue weight.

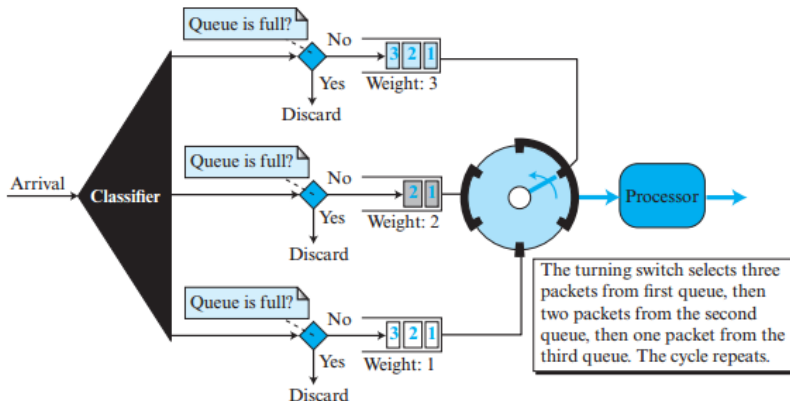
Weights $3 : 2 : 1 \Rightarrow 3, 2, 1$ packets per round

Illustration

For throughput R , the text gives illustrative shares $R/2$, $R/3$, and $R/6$ for three classes when packet sizes are equal.

Figure 8.61 – Weighted Fair Queuing

Figure 8.61 *Weighted fair queuing*



Traffic Shaping or Policing

- **Traffic shaping:** controls the amount/rate of traffic leaving a network.
- **Traffic policing:** controls traffic entering a network.
- Two techniques:
 - ① Leaky bucket
 - ② Token bucket

Leaky Bucket – Basic Idea

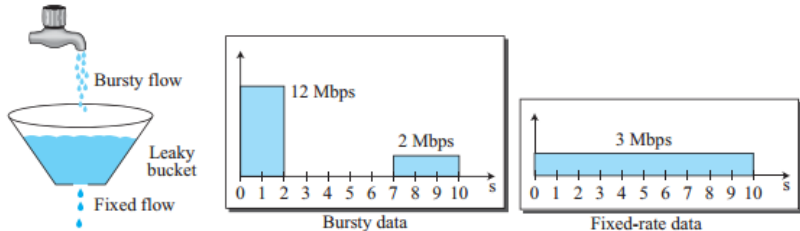
- Like water leaking through a small hole.
- Output rate remains constant while the bucket contains data.
- Input rate may vary.
- Bursty input is smoothed into controlled output.

Example

Committed bandwidth = 3 Mbps.
Burst: 12 Mbps for 2 s = 24 Mb.
Later: 2 Mbps for 3 s = 6 Mb.
Total = 30 Mb in 10 s.
Leaky bucket sends 3 Mbps over the 10 s period.

Figure 8.62 – Leaky Bucket and Traffic Smoothing

Figure 8.62 *Leaky bucket*



Leaky Bucket – Algorithm

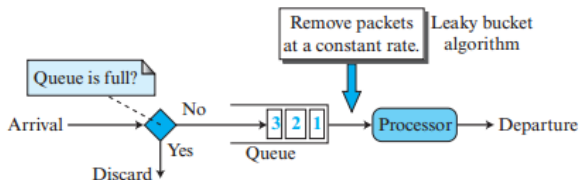
- 1 Initialize a counter to n at the clock tick.
- 2 If n is greater than the packet size, send the packet and decrement the counter by the packet size.
- 3 Repeat until the counter is smaller than the packet size.
- 4 Reset the counter to n and repeat at the next clock tick.

Variable-Length Packets

The fixed output rate is based on the number of bytes or bits rather than a fixed number of packets.

Figure 8.63 – Leaky Bucket Implementation

Figure 8.63 *Leaky bucket implementation*



Token Bucket

- Leaky bucket does not credit an idle host.
- Token bucket allows an idle host to accumulate credit for future bursts.
- Bucket capacity = c tokens.
- Tokens enter at r tokens/second.
- One token is removed for each cell/unit of data sent.
- Token bucket therefore permits controlled bursts while regulating the average rate.

Token Bucket – Example 8.12

Given

Bucket capacity = 10,000 tokens

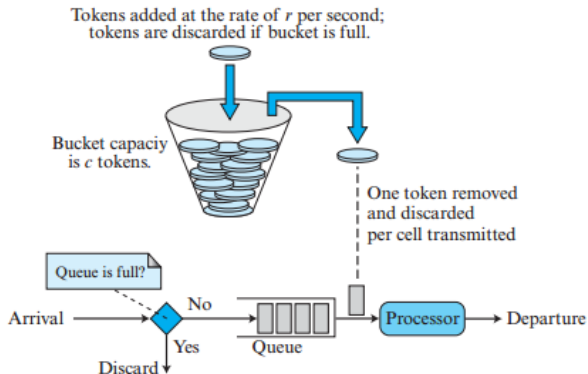
Token generation rate = 1,000 tokens/s

- After 10 seconds of inactivity, the bucket becomes full.
- Additional tokens are discarded when the bucket is full.

$$\text{Maximum average rate} = \frac{1000t + 10,000}{t}$$

Figure 8.64 – Token Bucket

Figure 8.64 *Token bucket*



Combining Token Bucket and Leaky Bucket

- The two techniques can be combined.
- Token bucket first gives credit to an idle host and regulates traffic.
- Leaky bucket is then applied to smooth the traffic.
- The leaky-bucket rate needs to be higher than the rate at which tokens are dropped into the bucket.

Resource Reservation

- A flow may need buffer, bandwidth, CPU time, etc.
- QoS improves when resources are reserved beforehand.

Admission Control

- Router/switch accepts or rejects a flow using flow specifications.
- Considers bandwidth, buffer, CPU capacity, and previous commitments.
- ATM calls this Connection Admission Control (CAC).

8.5.4 Integrated Services (IntServ)

- **IntServ** is a flow-based QoS model developed by IETF.
- Resources such as bandwidth are explicitly reserved for a particular flow.
- Three basic schemes:
 - ① Classify packets according to required service.
 - ② Use scheduling based on flow characteristics.
 - ③ Use admission control before committing resources.
- RSVP supplies the connection-oriented behavior needed over IP.

IntServ Flow Specification

Rspec – Resource Specification

Defines resources the flow needs to reserve, such as buffer and bandwidth.

Tspec – Traffic Specification

Defines the traffic characterization of the flow.

- Router decides to admit or deny the service based on current resources and previous commitments.

Guaranteed Service

- For real-time traffic requiring guaranteed minimum end-to-end delay.
- Guarantees delivery within a certain time when traffic remains within T_{spec} .
- Quantitative service.
- Example: Voice over IP.

Controlled-Load Service

- For applications tolerant of some delay but sensitive to overload and packet loss.
- Qualitative service.
- Examples: file transfer, e-mail, Internet access.

Resource Reservation Protocol (RSVP)

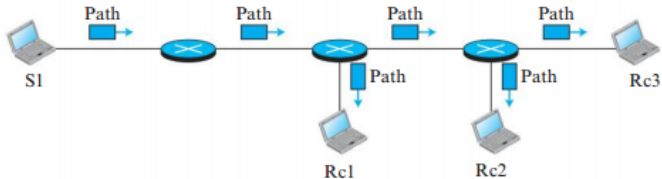
- IntServ needs connection-oriented network-layer behavior, while IP is connectionless.
- **RSVP** is an independent protocol running over IP to support resource reservation.
- RSVP is based on multicast communication but can also support unicast.
- Unicast is treated as a special case with one multicast-group member.

RSVP – Receiver-Based Reservation

- In RSVP, **receivers make the reservation**, not senders.
- This matches multicast routing, where receivers decide whether to join or leave.
- Two important RSVP messages:
 - 1 **Path**
 - 2 **Resv**

Figure 8.65 – RSVP Path Messages

Figure 8.65 *Path messages*

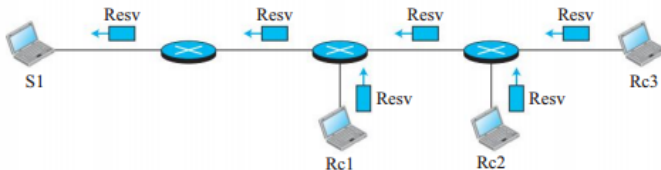


RSVP Path Messages

- Path travels from sender toward all receivers.
- It stores necessary path information for receivers.
- In multicast, a new Path message is created when the path diverges.

Figure 8.66 – RSVP Resv Messages

Figure 8.66 *Resv messages*



RSVP Resv Messages

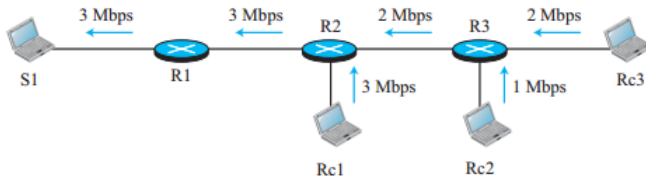
- After receiving Path, a receiver sends a **Resv message**.
- Resv travels upstream toward the sender.
- It reserves resources on routers supporting RSVP.
- A router without RSVP uses best-effort forwarding.

Reservation Merging

- RSVP reservations are merged rather than reserved separately for every receiver.
- Example:
 - Rc3 requests 2 Mbps.
 - Rc2 requests 1 Mbps.
 - R3 reserves 2 Mbps, the larger request.
- Different receivers may request different multimedia quality levels.

Figure 8.67 – RSVP Reservation Merging

Figure 8.67 *Reservation merging*



RSVP Reservation Styles

- ➊ **Wildcard Filter (WF):** one reservation for all senders, based on the largest request.
- ➋ **Fixed Filter (FF):** distinct reservation for each flow.
- ➌ **Shared Explicit (SE):** one reservation shared by a specified set of flows.

Use

WF is suited to non-simultaneous sender flows; FF to potentially simultaneous flows; SE shares a reservation among specified flows.

- Reservation information stored at each node must be refreshed periodically.
- This is called a **soft state**.
- Hard-state protocols retain flow information until explicitly erased.
- The default refresh interval stated in the text is **30 seconds**.

Problems with Integrated Services

Scalability

- Every router keeps information for every flow.
- Becomes difficult as the Internet grows.
- Especially troublesome for high-speed core routers.

Service-Type Limitation

- Only two service types are defined:
- Guaranteed
- Controlled-load
- Applications may need more service types.

8.5.5 Differentiated Services (DiffServ)

- DiffServ addresses shortcomings of Integrated Services.
- Applications mark packets into classes according to priority.
- Routers and switches use queuing strategies to handle the classes.

Two Fundamental Changes

- 1 Main processing moves from network core to network edge.
- 2 Per-flow service changes to **per-class service**.

IntServ vs DiffServ

	IntServ	DiffServ
Service basis	Per flow	Per class
Processing	Flow accommodations	Class marking/treatment
Router information	Per-flow state	Class/PHB information
Scalability concern	High per-flow state	Reduced core state
Service definition	Two defined classes	Multiple classes can be defined

- Each DiffServ packet contains a **DS field**.
- The field is set at the network boundary by the host or boundary router.
- IETF proposes replacing IPv4 ToS or IPv6 priority class with the DS field.

DS Subfields

- **DSCP**: 6 bits; defines Per-Hop Behavior (PHB).
- **CU**: 2 bits; Currently Unused.

Per-Hop Behavior (PHB)

- ① **DE PHB – Default:** same as best-effort delivery.
- ② **EF PHB – Expedited Forwarding:**
 - Low loss
 - Low latency
 - Ensured bandwidth
- ③ **AF PHB – Assured Forwarding:**
 - High assurance when traffic stays within the traffic profile.
 - Some packets may still be discarded.

Figure 8.68 – DS Field

Figure 8.68 *DS field*



- DiffServ uses traffic conditioners such as:

- 1 **Meter**
- 2 **Marker**
- 3 **Shaper**
- 4 **Dropper**



Traffic Conditioner – Components

Meter

- Checks whether traffic matches the negotiated profile.
- Can use a token bucket.

Marker

- Can remark or down-mark packets.
- Does not up-mark a packet.

Shaper

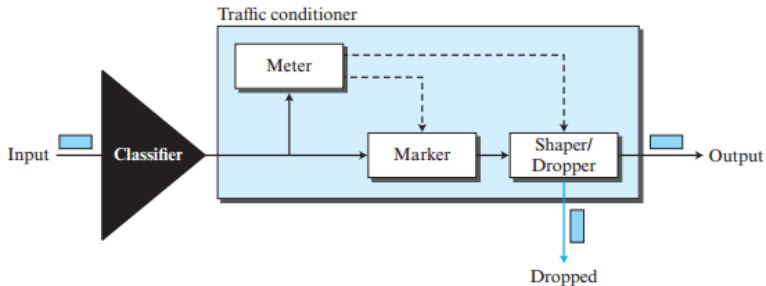
- Reshapes non-compliant traffic.

Dropper

- Shaper without a buffer.
- Discards packets when traffic severely violates the profile.

Figure 8.69 – DiffServ Traffic Conditioner

Figure 8.69 Traffic conditioner



8.5 – Complete QoS Picture

- ① Flow characteristics: reliability, delay, jitter, bandwidth.
- ② Scheduling: FIFO, priority queuing, weighted fair queuing.
- ③ Traffic shaping/policing: leaky bucket and token bucket.
- ④ Resource reservation and admission control.
- ⑤ IntServ: flow-based reservation with RSVP.
- ⑥ DiffServ: class-based treatment using DSCP and PHBs.

Key Exam Points

- QoS characteristics: **reliability, delay, jitter, bandwidth.**
- FIFO: no service differentiation.
- Priority queuing: possible **starvation.**
- WFQ: weighted, fair service among queues.
- Leaky bucket: smooths bursty traffic.
- Token bucket: allows controlled bursts and credits idle periods.
- IntServ: **flow-based**; RSVP supports resource reservation.
- IntServ classes: **Guaranteed** and **Controlled-load.**
- DiffServ: **per-class**; DSCP defines PHB.
- PHBs: **DE, EF, AF.**
- Traffic conditioner: **meter, marker, shaper, dropper.**